SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID –** 25019
- **Problem Statement Title- Digital Learning Platform for Rural School Students in Nabha**
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID- 67676**
- **Team Name - MindMatrix**

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Gyan-Kiran: The Offline-First Learning Platform

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PRESENT SYSTEM: The current education system in rural Nabha struggles with **outdated computer infrastructure** and **unreliable internet connectivity.** Teachers and students cannot effectively use digital tools and access to modern educational content is **limited.** This results in **low digital literacy** and creates a significant gap between rural and urban education standards.

PROPOSED SOLUTION: Our Gyan-Kiran Platform transforms education for rural schools by providing a specialized, **fully offline-capable** mobile and web application. It features **downloadable interactive lessons** in local languages, **simplified UI experience** and **centralized teacher dashboards** to track **student progress**, ensuring **consistent** and **effective learning** even with poor or no connectivity.

Process flow of the proposed system

Teacher Creates Content

Teacher prepares course material offline. Content is saved locally and queued for automatic upload.

→

Sync & Learn Offline

Students download the course once. The app then intelligently syncs only new or updated content when online.

→

Auto-Sync Progress

Student progress and quiz results are saved instantly on the device. All data is automatically synced to the backend when a connection is available.

→

Track & Analyze Performance

The teacher's dashboard works offline using the last synced data. It updates with the latest class performance when online.

Key Features:

- **True Offline-First Architecture:** A robust **Sync Manager** allows students to complete lessons and take quizzes entirely **offline**. Progress is automatically synced to the cloud whenever a connection is available.
- **Secure & Private Classrooms:** Each school gets a dedicated, secure space managed by teachers via a **unique School Invite Code**, ensuring complete data privacy.
- **Insightful Teacher Dashboards:** A simple web-based dashboard providing teachers with **real-time analytics** on student performance and class-wide trends.
- **Hyper-Localized & Accessible:** The entire platform will be available in **Punjabi**, with accessibility features like **audio-assisted** lessons for students with reading difficulties.
- **AI-Powered Help Assistant:** An integrated **chatbot** provides students with instant, 24/7 answers to common questions, reducing teacher workload.

TECHNICAL APPROACH

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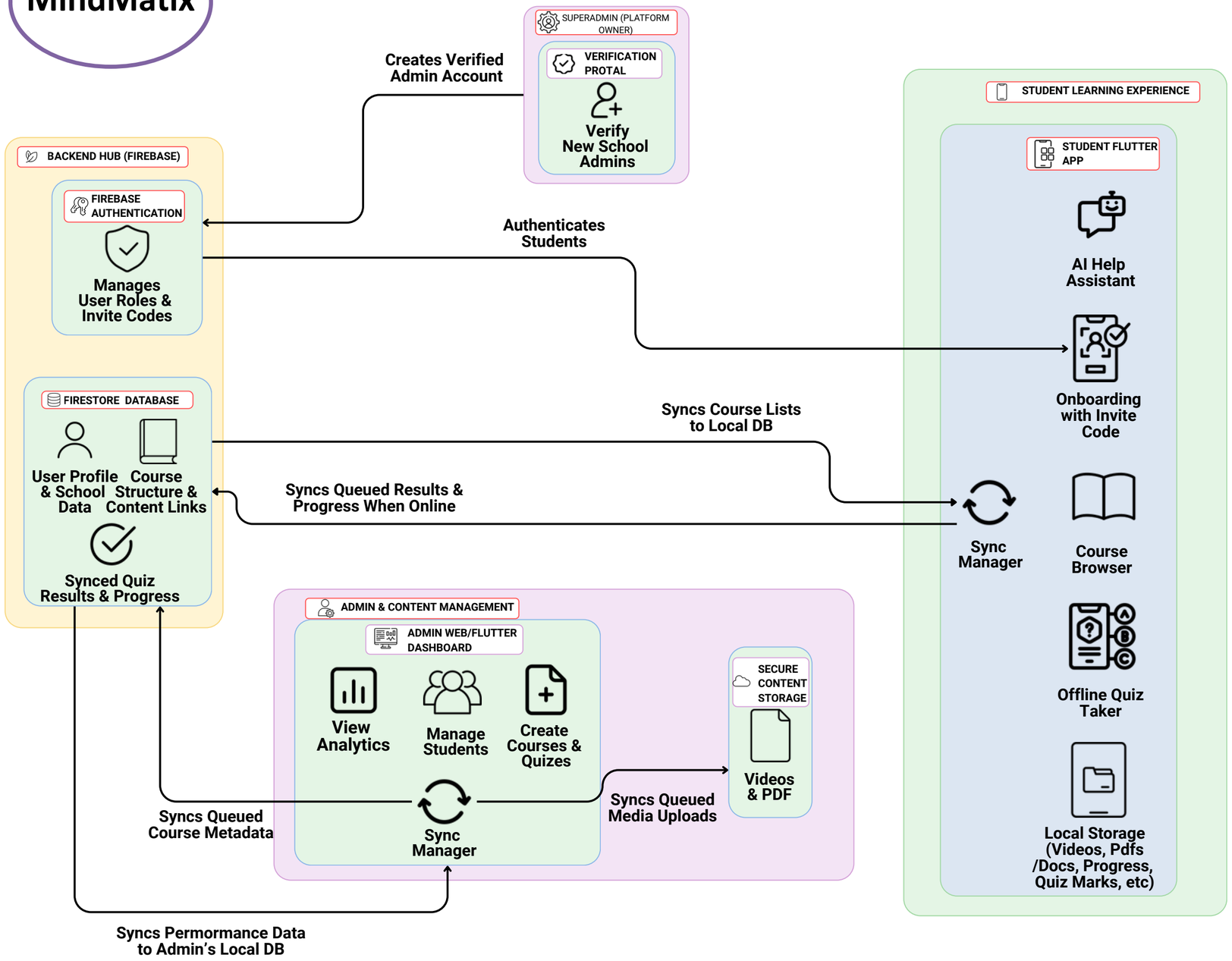

Scalable Multi-School Architecture

- Our platform is built for **multiple organizations**. Each organization is treated as a secure and private **"tenant"** to ensure data privacy.
- **Teacher Verification Process:**
  a. A new teacher **requests** an account via a simple sign-up form.
  b. Our team **(SuperAdmins)** verifies the teacher's **credentials** through a manual check (e.g., school website, official email).
  c. Once approved, the teacher's account is created as an **"Admin."**
- The verified **Admin** receives a **unique School ID** and a shareable **Invite Code** for their **private digital classroom.**
- Students use this **Invite Code** upon **registration** to join their **specific organization**, ensuring they only see relevant courses.
- **Firebase Security Rules** enforce strict data isolation, making one school's data completely invisible to another.

Technology Stack:

- **Frontend (Mobile & Web): Flutter** For a single, high-performance codebase that runs natively on **Android** and the **Web**.
- **Backend & Database:** Firebase Using **Firestore** for **real-time data** (courses, quiz results) and **Firebase Authentication** for secure, role-based user management.
- **Secure Content Delivery: Telegram / Cloud Storage** For hosting and delivering **downloadable video** and **PDF** content securely to the app.
- **Local Storage: SQLite / Hive (Flutter)** To cache downloaded course materials (videos, notes) for a seamless **offline learning experience**.
- **AI Help Assistant: Google AI Platform / Gemini API** To power the integrated chatbot for instant student query resolution.

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FEASIBILITY AND VIABILITY

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Potential Challenges and Risks

Technical:

•**Offline Data Synchronization:** Architecting a robust sync mechanism between the local device and the Firebase backend is complex, especially with existing data structures like the rating system.

•**Device Storage Management:** Downloaded courses can consume significant space on the low-end smartphones common in rural areas.

User Adoption:

• **Teacher Onboarding:** Teachers in rural areas may have low digital literacy and could be resistant to adopting new technology.

•**Student Engagement:** Keeping students motivated to use a learning app for self-study outside of a structured classroom environment.

Strategies for Overcoming Challenges

Technical Solutions:

• **Phased Offline Rollout:** Initially implement a core "Download Video" feature. Post-selection, architect a dedicated sync module using Firebase's offline persistence to handle complex course data.

• **Smart Storage Manager:** Implement features to allow students to archive completed chapters and provide clear data on storage usage.

User Adoption:

•**Intuitive UI & Workshops:** Design an incredibly simple teacher dashboard and conduct on-site training workshops in Nabha.

•**Gamification:** Integrate points, badges, and leaderboards to boost student motivation and make learning more engaging.

Feasibility Analysis

Technological Feasibility:

• **Proven Core Framework:** The solution is built upon a functional application using Flutter and Firebase, which serves as a powerful Proof of Concept.

•**Scalable Backend:** Utilizing Firebase provides a scalable, serverless architecture for user management, authentication, and data storage.

Educational Impact:

•**Bridging the Digital Divide:** Directly addresses the lack of digital resources in Nabha by providing accessible educational content, even without consistent internet.

• **Data-Driven Insights:** Empowers teachers with valuable data to track student performance and provide personalized support.

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IMPACT AND BENEFITS

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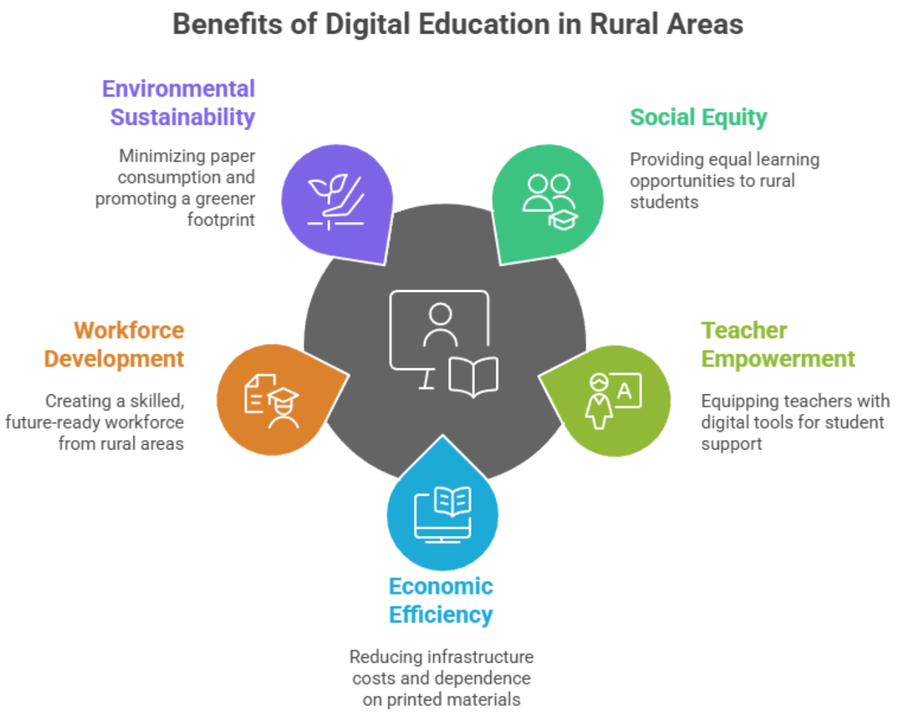

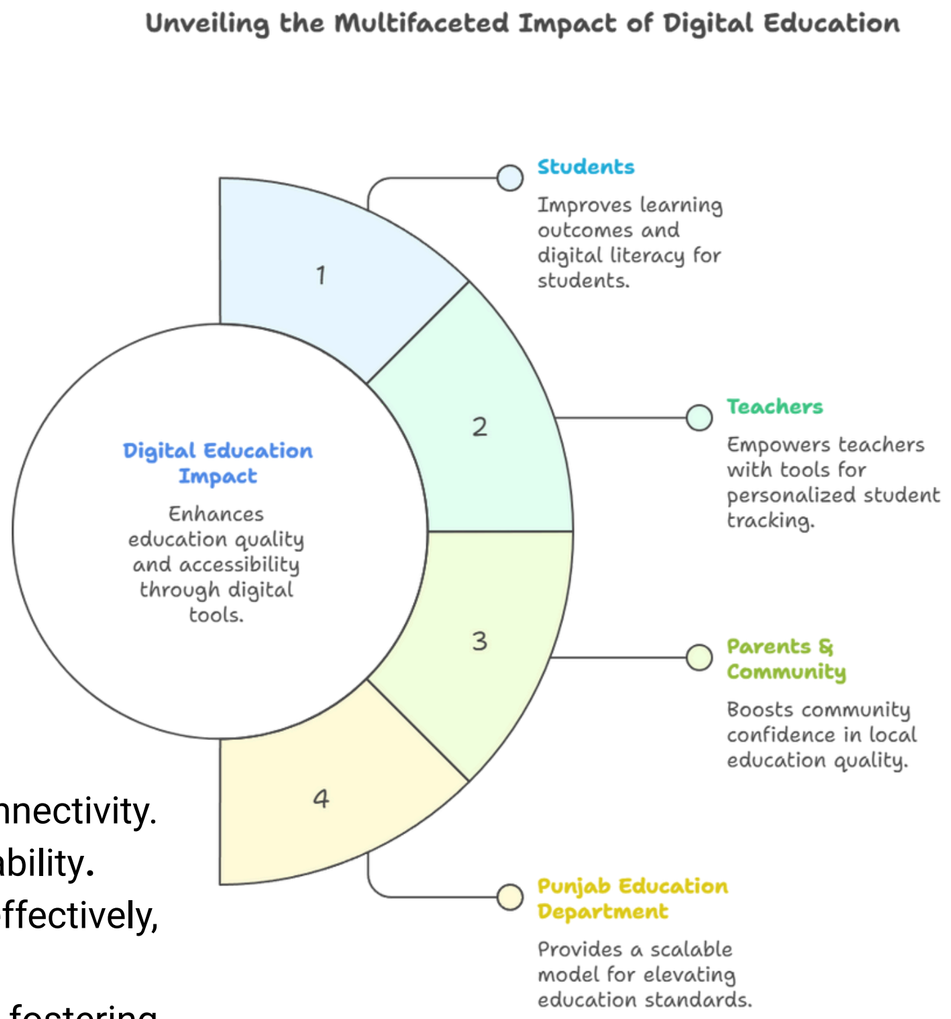

Potential Impact on Target Audience:

- **Students:** Will gain access to quality digital education, regardless of their internet connectivity. This will improve learning outcomes, boost digital literacy, and enhance future employability.
- **Teachers:** Will be empowered with digital tools to track student performance effectively, allowing them to provide personalized attention and reduce manual record-keeping.
- **Parents & Community:** Will see a tangible improvement in the quality of education, fostering community confidence in the local school system.
- **Punjab Education Department:** Will gain a scalable, data-driven model to elevate education.

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RESEARCH AND REFERENCES

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The Problem in Numbers: The Digital Disconnect in Rural Punjab

- **The Core Paradox:** According to the ASER 2022 Report, while over 85% of children in rural Punjab have access to a smartphone, official government data from UDISE+ reports that only a fraction of schools have functional internet, creating a critical infrastructure gap our offline-first solution targets.
- **The Learning Gap:** The ASER report also highlights a significant learning gap, showing that a large percentage of students in Standard III cannot read a Standard II text, proving the need for supplemental learning tools to reinforce foundational skills.

Table 6: Trends over time
Reading in Std V and Std VIII. By school type.
2014, 2016, 2018, 2022, 2024

Year	% Children in Std V who can read Std II level text			% Children in Std VIII who can read Std II level text		
	Govt	Pvt	Govt & Pvt*	Govt	Pvt	Govt & Pvt*
2014	60.9	73.8	66.6	87.3	84.4	86.2
2016	64.0	73.8	69.1	83.6	90.0	86.3
2018	68.7	74.4	71.6	83.8	87.1	85.1
2022	59.4	75.5	66.2	82.6	90.2	85.4
2024	60.8	62.2	61.4	72.2	84.6	76.7

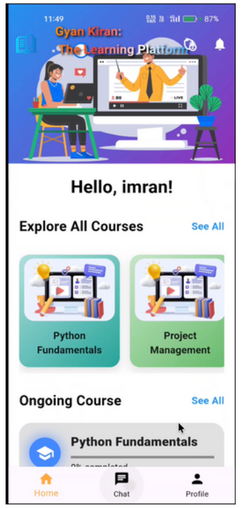
The Competitive Gap: Why Existing Solutions Fail

- **Government Platforms (DIKSHA, ePathshala):** An analysis of Google Play Store user reviews reveals critical reliability issues, with widespread complaints of non-functional offline modes and buggy interfaces that hinder adoption.
- **Private EdTech (e.g., Khan Academy):** Our competitive analysis shows that while these platforms offer quality content, they lack hyper-localization for the Punjabi curriculum and, crucially, do not provide an integrated dashboard for teachers to manage their specific classrooms.

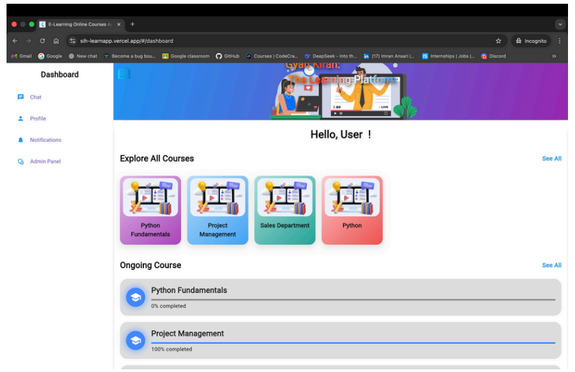
Table 10: Smartphone availability and use. By age. 2024

Age	% Children who:			Of those who can use a smartphone, % who have their own smartphone
	Have a smartphone at home	Could bring a smartphone to do digital tasks*	Can use a smartphone	
14	96.8	77.3	94.0	38.3
15	95.6	79.9	94.1	46.8
16	95.9	82.1	94.6	56.8
All	96.2	79.4	94.2	46.0

Demo(MVP) and Prototype Links



Mobile (User dashboad)



Website (User dashboad)

- Web site link please open in incognito mode: https://sih-learnapp.vercel.app/
- Mobile Application base apk file : https://drive.google.com/file/d/15OEGyDtFjvw8jltrQp6UAglKvD_36lYr/view
- Youtube link: https://www.youtube.com/watch?v=tpfMCxDsdGk

Sources & References

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- The Tribune: "In rural Punjab, 85% kids have access to smartphones" https://www.tribuneindia.com/news/punjab/in-rural-punjab-85-kids-have-access-to-smartphones/
- Competitive Analysis (e.g., Khan Academy) https://play.google.com/store/apps/details?id=org.khanacademy.android